THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Three identical-looking bars P, Q, R: P attracts Q at both ends, Q attracts R at both ends, R repels P at one end. Which can be concluded? 1

(a) P and R are magnets, Q is magnetic material
(b) all are iron
(c) all are magnets
(d) all are plastic

2. Which of the following is the best argument for using a magnet to separate iron from copper turnings? 1

(a) iron is magnetic and copper is not **(b)** both are shiny
(c) both are metals **(d)** both rust

3. An iron nail is rubbed with the north pole of a magnet in one direction many times. The end where the stroke ends becomes a: 1

(a) neutral **(b)** no pole **(c)** south pole **(d)** north pole

4. A freely hung bar magnet has one end painted red; the red end settles pointing towards the geographic south. The red end is the: 1

(a) north-seeking pole **(b)** south-seeking pole
(c) non-magnetic part **(d)** neutral centre

5. A mixture contains aluminium, iron and wooden pieces. A magnet separates: 1

(a) iron only **(b)** all metals
(c) iron and aluminium **(d)** wood

6. A student suspends magnet A and brings magnet B's north pole near A's north-seeking end. A turns away. This shows: 1

(a) A is iron **(b)** B has no poles
(c) B's north attracts A's north **(d)** B's north repels A's north

7. Two magnets, each with poles marked, are placed in a line with N of A facing S of B and they attract. If B is reversed, they will: 1

(a) melt **(b)** stay same **(c)** repel **(d)** attract more

8. Why does a steel almirah become slightly magnetic over many years? 1

(a) it gets heated **(b)** it sees the sun
(c) Earth's field aligns the tiny magnets **(d)** it rusts

9. If the Earth's geographic north pole region behaves like a magnetic south pole, a compass's north needle is attracted to it because: 1

(a) wind **(b)** like poles attract
(c) gravity **(d)** unlike poles attract

10. A compass kept on a table shows north in a wrong direction. The best explanation is: 1

(a) sunlight **(b)** a magnet or iron object nearby
(c) the table is wooden **(d)** noise

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A compass is placed on a steel table and gives a wrong reading. Explain and suggest a remedy. 2

12. A student says that heating a magnet strengthens it. Do you agree? 2

13. Why do we say 'magnets act through non-magnetic materials'? 2

14. Why does an iron nail dangling from a magnet drop other nails when the magnet is pulled away? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how to show that the poles of a magnet are not at the very ends but near them, using iron filings. 3

16. Explain how a loudspeaker uses a magnet. 3

17. Explain the term 'magnetic field'. 3

OR

Explain how the magnetic compass helped in exploring new lands. 3

18. How can you show the Earth's magnetism with a simple experiment? 3

OR

Describe the effect of a steel sheet kept between a magnet and an iron nail. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an investigation to compare the strength of two magnets using paper clips and describe how to make it a fair test. 5

OR

Describe an activity to show that the strength of a magnet is highest at the poles, and write observations and a conclusion. 5

20. A box has three unlabelled bars: a magnet, an iron bar and a copper bar. Describe a method with only the bars to identify each. 5

OR

Analyse why magnets lose strength and design a storage plan. 5

END OF QUESTION PAPER — Total: 40 Marks